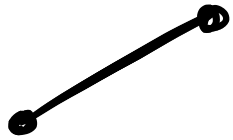
Arc length of curves

Basic skills

Q1. (x_1, y_1) to (x_2, y_2)  is Pythagoras

so $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Q2. $y = 2x$ $\quad (0,0)$ to $(5, 10)$

a) $d = \sqrt{5^2 + 10^2} = \sqrt{125} = 5\sqrt{5}$

b) $\frac{dy}{dx} = 2$ so $\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{1 + 2^2} = \sqrt{5}$

so length $= \int_0^5 \sqrt{5}\, dx = \left[\sqrt{5}x\right]_0^5 = 5\sqrt{5}$

same as a)

Q3. $x = t^2,\ y = t^3$

a) $\frac{dx}{dt} = 2t \quad \frac{dy}{dt} = 3t^2 \quad \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = \sqrt{4t^2 + 9t^4}$

so length $= \int_0^1 \sqrt{4t^2 + 9t^4}\, dt$

b) $= \int_0^1 t\sqrt{9t^2+4}\,dt$ let $u = 9t^2+4$

$\frac{du}{dt} = 18t$

$= \int_0^1 \cancel{t}\sqrt{u}\ \frac{1}{18\cancel{t}}\,du$

$= \frac{1}{18}\left[\frac{2}{3}u^{3/2}\right]_0^1 = \frac{1}{18}\left(\frac{2}{3}-0\right) = \frac{1}{27}$

Competency skills

Q4. $x = 2t^2+1,\ y = 4t \qquad 0 \leq t \leq 2$

a) $\frac{dx}{dt} = 4t \quad \frac{dy}{dt} = 4$ so $\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2} = \sqrt{16t^2+16}$

so length $= \int_0^2 \sqrt{16t^2+16}\,dt = \int_0^2 4\sqrt{t^2+1}\,dt$

b) let $t = \sinh u$ so $t^2+1 = \cosh^2 u$

$\frac{dt}{du} = \cosh u$

so $\int_0^{\sinh^{-1}2} 4\sqrt{\cosh^2 u}\ \cosh u\,du$

$= \int_0^{\sinh^{-1}2} 4\cosh^2 u\,du$ $\qquad \cosh 2x = 2\cosh^2 x - 1$

$= \int_0^{\sinh^{-1}2} 4\left(\frac{1}{2}+\frac{1}{2}\cosh 2u\right)du$

$$= \left[2u + \sinh 2u \right]_0^{\sinh^{-1} 2}$$

$\downarrow$ $2\sinh u \cosh u$

$$= \left[2u + 2\sqrt{\sinh^2 u + 1}\ \sinh u \right]_0^{\sinh^{-1} 2}$$

$$= 2\sinh^{-1} 2 + 2\sqrt{2^2 + 1} \times 2$$

$$= 2\sin^{-1} 2 + 4\sqrt{5}$$

Q5. $y = \cosh x \qquad \dfrac{dy}{dx} = \sinh x$

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{1 + \sinh^2 x} = \sqrt{\cosh^2 x} = \cosh x$$

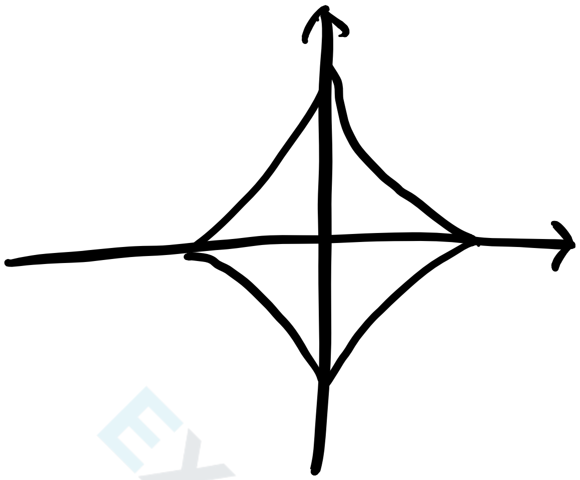
So length is $\displaystyle\int_0^{\ln a} \cosh x \, dx = \left[\sinh x \right]_0^{\ln a}$

$$= \sinh(\ln a) - 0 = \frac{e^{\ln a} - e^{-\ln a}}{2} - 0$$

$$= \frac{a - \frac{1}{a}}{2} = \frac{a^2 - 1}{2a}$$

Q6. $x = a\cos^3\theta,\ y = a\sin^3\theta$

a)



"collapsed circle"

b) $\frac{dx}{d\theta} = -3a\cos^2\theta\sin\theta \qquad \frac{dy}{d\theta} = 3a\sin^2\theta\cos\theta$

so $\sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} = \sqrt{9a^2\cos^4\theta\sin^2\theta + 9a^2\sin^4\theta\cos^2\theta}$

$= 3a\cos\theta\sin\theta\sqrt{\cos^2\theta + \sin^2\theta} = 3a\cos\theta\sin\theta$

$= \frac{3}{2}a\sin 2\theta$

limits $\frac{\pi}{2}$ as do first quad then $\times 4$

length of 1st quad $= \int_0^{\frac{\pi}{2}} \frac{3}{2}a\sin 2\theta \, d\theta = \left[-\frac{3}{4}a\cos 2\theta\right]_0^{\frac{\pi}{2}}$

$= -\frac{3}{4}a \times -1 \ - \ -\frac{3}{4}a = \frac{3}{2}a$

so for all 4 have $\frac{3}{2}a \times 4 = 6a$

Q7. $\displaystyle\int_0^k \underbrace{\sqrt{1+(e^x)^2}}_{\downarrow \text{ int}}\, dx = 1$

$\displaystyle\int \sqrt{1+e^{2x}}\, dx$

now let
$u = e^{2x}$
$du = 2e^{2x}\, dx$
$= 2u\, dx$

$\Rightarrow \displaystyle\int \sqrt{1+u} \times \frac{1}{2u}\, du$

let $v = \sqrt{1+u}$
$\frac{dv}{du} = \frac{1}{2\sqrt{1+u}}$
$du = 2v\, dv$

$\Rightarrow \displaystyle\int \frac{1}{2} v \times \frac{1}{v^2-1} \times 2v\, dv$

$\Rightarrow \displaystyle\int \frac{v^2}{v^2-1}\, dv$

Partial fractions $\frac{v^2}{v^2-1} = A + \frac{B}{v+1} \quad \frac{C}{v-1}$

$\Rightarrow A(v^2-1) + B(v-1) + C(v+1) = v^2$

$\Rightarrow A = 1,\ B + C = 0,\ -A - B + C = 0$

$\Rightarrow -B + C = 1 \quad \Rightarrow C = \frac{1}{2},\ B = -\frac{1}{2}$

so $\displaystyle\int 1 + \frac{1}{2}\left(\frac{1}{v-1} - \frac{1}{v+1}\right) dv$

$$= v + \frac{1}{2}\ln(v-1) - \frac{1}{2}\ln(v+1)$$

$$= v + \frac{1}{2}\ln\left(\frac{v-1}{v+1}\right)$$

$$= \sqrt{1+u} + \frac{1}{2}\ln\left(\frac{\sqrt{1+u}-1}{\sqrt{1+u}+1}\right)$$

$$= \sqrt{1+e^{2x}} - \frac{1}{2}\ln\left(\frac{\sqrt{1+e^{2x}}+1}{\sqrt{1+e^{2x}}-1}\right)$$

$$= \sqrt{1+e^{2x}} - \tanh^{-1}\left(\sqrt{1+e^{2x}}\right)$$

so length is

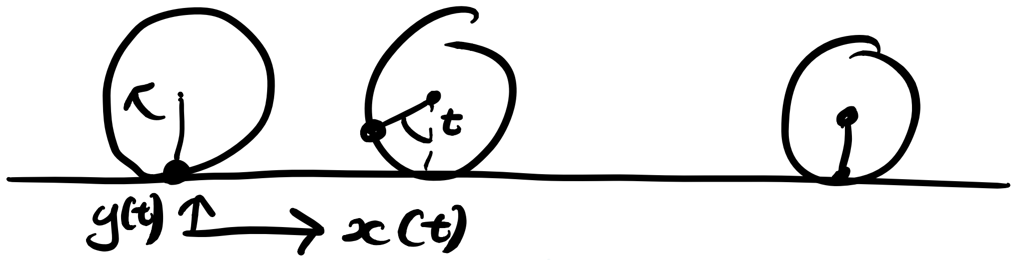
$$\left(\sqrt{1+e^{2k}} - \tanh^{-1}\left(\sqrt{1+e^{2k}}\right)\right) - \left(\sqrt{2} - \tanh^{-1}(\sqrt{2}\right) = 1$$

so $\sqrt{1+e^{2k}} - \tanh^{-1}\left(\sqrt{1+e^{2k}}\right) = 1 + \sqrt{2} - \tanh^{-1}(\sqrt{2})$

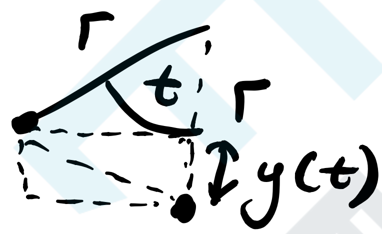
Hard to solve but we only need an equation for k

Advanced skills

Q8. a)



so

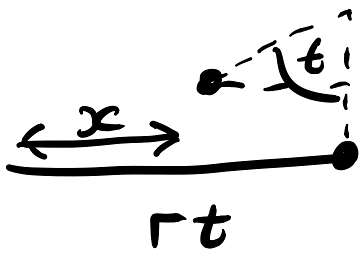


So height is $y(t) = r - r\cos t$

$= r(1 - \cos t)$

For traveled distance arcl length is rt

now that is along ground



so $x(t) = rt - r\sin t$

$= r(t - \sin t)$

b) $\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = \sqrt{(r(1-\cos t))^2 + (r\sin t)^2}$

$= \sqrt{r^2(1-\cos t)^2 + r^2\sin^2 t} = r\sqrt{1 - 2\cos t + \cos^2 t + \sin^2 t}$

$$= r\sqrt{2-2\cos t} = \sqrt{2}\, r\sqrt{1-\cos t}$$

so $\displaystyle\int_0^{2\pi} \sqrt{2}\, r\sqrt{1-\cos t}\; dt$

As $\cos 2\theta = 1 - 2\sin^2\theta$

$\sqrt{1-\cos 2\theta} = \sqrt{2}\sin\theta$

let $2\theta = t$

$$\int_0^{2\pi} \sqrt{2}\, r\, \sqrt{2}\sin\left(\tfrac{t}{2}\right) dt$$

$$= \left[-2r\cos\left(\tfrac{t}{2}\right)\times 2\right]_0^{2\pi} = \left[-4r\cos\left(\tfrac{t}{2}\right)\right]_0^{2\pi}$$

$$= -4r\times -1 - -4r\times 1 = 8r$$

Q9. $x = a\cos\theta$, $y = b\sin\theta$, $e = \dfrac{\sqrt{a^2-b^2}}{a}$

a) $\dfrac{dx}{d\theta} = -a\sin\theta \qquad \dfrac{dy}{d\theta} = b\cos\theta$

$b^2 = (1-e^2)a^2$

$$\sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} = \sqrt{a^2\sin^2\theta + b^2\cos^2\theta}$$

$$= a\sqrt{\sin^2\theta + \frac{b^2}{a^2}\cos^2\theta} = a\sqrt{\sin^2\theta + \frac{(1-e^2)a^2}{a^2}\cos^2\theta}$$

$$= a\sqrt{\sin^2\theta + \cos^2\theta - e^2\cos^2\theta} = a\sqrt{1-e^2\cos^2\theta}$$

so as 4 equal quadrants

$$4\int_0^{\frac{\pi}{2}} a\sqrt{1-e^2\cos^2\theta}\, d\theta = 4a\int_0^{\frac{\pi}{2}}\sqrt{1-e^2\cos^2\theta}\, d\theta$$

b)

$$\approx 4a\int_0^{\frac{\pi}{2}} 1-\tfrac{1}{2}e^2\cos^2\theta \quad d\theta$$

use $(1+ax)^n \approx 1+nax+\ldots$ expansion as e small

$$= 4a\int_0^{\frac{\pi}{2}} 1-\tfrac{1}{2}e^2\left(\tfrac{1}{2}+\tfrac{1}{2}\cos 2\theta\right) d\theta$$

$$= 4a\left[\left(1-\tfrac{1}{4}e^2\right)\theta + \frac{e^2}{8}\sin 2\theta\right]_0^{\frac{\pi}{2}}$$

$$= 4a\left(\left(1-\tfrac{1}{4}e^2\right)\frac{\pi}{2} + 0 - (0+0)\right)$$

$$= 2\pi a\left(1-\tfrac{1}{4}e^2\right)$$